

Unit tests

Why were these tests chosen and why the tested code is critical 35 tests across 5 files

5 Test Files

- AuthServiceImplTest.java
- DataAccessExceptionAspectTest.java
- TourMetricsCalculatorTest.java
- TourServiceImplOwnershipTest.java
- TourServiceSearchTest.java

AuthServiceImpl

Tests login and authentication logic. ---> Critical because incorrect auth handling could allow unauthorized access or incorrectly reject valid users.

DataAccessExceptionAspectTest

Tests that database exceptions are correctly caught and wrapped into layer-appropriate exceptions. ---> Critical because without this, raw persistence exceptions would leak into the presentation layer, violating the layered architecture requirement.

TourMetricsCalculatorTest

Tests the computed attributes (popularity, child-friendliness). ---> Critical because these values are derived from log data and displayed to the user — wrong formulas would silently show incorrect information.

TourServiceImplOwnershipTest.java

Tests that users can only access/modify their own tours. ---> Critical because without ownership checks, any logged-in user could read, edit or delete another user's data — a serious security vulnerability.

TourServiceSearchTest.java

Tests the full-text search and sorting functionality across tour and log fields including computed attributes. ---> Critical because search is a core feature and incorrect filtering could show wrong results or miss data.